On the Impossibility of Traversing Certain Configurations n_3 in a Closed Path

By *Ernst Steinitz* in Berlin.

The definition usually given for the configurations n_3 allows one to regard two configurations Cf. m_3 and Cf. m'_3 as a single configuration Cf. $(m+m')_3$. We shall retain this convention and call a configuration Cf. n_3 which is composed in this way from two configurations *decomposable*.

The property of a configuration Cf. n_3 of being decomposable or not decomposable can also be presented somewhat differently. Each line of the configuration Cf. n_3 has three configuration-segments, or briefly "segments," by which are meant the joins of each two of its points. We call a polygonal line composed of such segments a "path," and we call two points "connected" if it is possible to get from one to the other along a path. It is then clear that two points which are connected with a third are also connected with each other. If we call a configuration Cf. n_3 connected when connectedness holds between all its points, then a decomposable configuration Cf. n_3 is certainly disconnected; but it is also easy to see that an indecomposable configuration Cf. n_3 is connected.

For $n = 7, 8, 9, 10$, Mr. *Schröter*[1] was probably the first to prove that every configuration Cf. n_3 can be traversed in a closed path and accordingly can be represented as an n-gon inscribed in and circumscribed about itself, whose vertices are the points of the configuration, while each of the n lines supplies one of its three segments as a side. For $n = 11$ I have succeeded in proving the same theorem; but in what follows I shall show that this theorem is not true for arbitrary n, as Mr. *S. Kantor*[2] thinks.

It is first evident that a decomposable configuration certainly cannot be traversed in one path; but even if one disregards this trivial truth, *Kantor*'s theorem remains false. To prove this, we begin with some auxiliary considerations.

Lemma. If a connected configuration Cf. n_3 *contains the configuration-line 1, 2, 3, then it is possible to get from every point of the configuration to the point 1 without using the segments 12 or 13. Equivalently: the connectedness of the figure is not destroyed if the segments 12 and 13 are removed.*

Proof. Suppose that there also exist points from which one cannot reach 1 without traversing one of the segments 21 or 31. Let the totality of these points be denoted by $\mathfrak{B}$, and the totality of the remaining points by $\mathfrak{A}$. Then the points 2 and 3 belong to $\mathfrak{B}$. Indeed, if b is any point of $\mathfrak{B}$, then, since the configuration Cf. n_3 is connected, there is a path leading from b to 1, and we may suppose that this path does not reach the point 1 at any time before its end. By assumption this path must contain one of the segments 21, 31, say 21. Since it reaches the point 1 only at its end, 21 must be the last segment of the path, and hence the part of the path leading from b to 2 contains neither of the segments 21, 31. If one of the points 2, 3, and therefore also the other (because they are joined by the segment 23), belonged to the totality $\mathfrak{A}$, then one could get from 2 to 1, and hence also from b to 1, along a path which contains neither of the segments 21, 31. This contradicts the assumption that b belongs to $\mathfrak{B}$. We now denote by A the totality of all lines which contain at least one point of $\mathfrak{A}$, but we exclude from this totality the line 1, 2, 3, because after the removal of the segments 21, 31 it really contains only the two points 2, 3 belonging to $\mathfrak{B}$. It is then immediate that not merely one point, but every point of any line of A belongs to $\mathfrak{A}$. If, therefore, we count how often it happens that a point of $\mathfrak{A}$ lies on a line of A, the number sought is

$$3g,$$

[1] Göttinger Nachrichten, 1889.
[2] Wiener Sitzungsberichte, vol. 84, II, p. 1305.

where g denotes the number of lines in A. The same number can also be determined in another way. Through every point of $\mathfrak{A}$ pass three lines which, by definition, belong to A; the only exception is the point 1, through which pass only two lines of A, since the third, namely 1, 2, 3, does not belong to A. Thus, if v is the number of points belonging to $\mathfrak{A}$, the number sought is

$$3v - 1,$$

and we obtain the equation

$$3g = 3v - 1,$$

which is plainly impossible. This eliminates the assumption that such points exist from which the point 1 cannot be reached without using one of the segments 21, 31.

Now let Cf. n_3 and Cf. m_3 be two fixed connected configurations. Let the first, whose points we denote by $1, 2, \ldots, n$, contain the line 1, 2, 3, and let the second, whose points we denote by $1', 2', \ldots, m'$, contain the line $1', 2', 3'$. Together, the two configurations may be regarded as a configuration Cf. $(m+n)_3$, which is of course not connected. If in the line 1, 2, 3 we remove the point 1 and replace it by $1'$, and in the line $1', 2', 3'$ we remove the point $1'$ and replace it by 1, then we plainly again have the scheme of a definite configuration Cf. $(m+n)_3$, which we shall denote by $\overline{\text{Cf.}\,(m+n)_3}$; and we shall now show that this configuration Cf. $(m+n)_3$ is connected.

Since the configuration Cf. n_3 is connected by assumption, one can get from each of its points to the point 1 along a path which contains neither of the segments 21, 31. The same path also exists within $\overline{\text{Cf.}\,(m+n)_3}$, because this configuration contains all segments of Cf. n_3 except 21 and 31, which have been replaced by $21'$ and $31'$. Hence within $\overline{\text{Cf.}\,(m+n)_3}$ all points denoted by unprimed numerals are connected with the point 1 and therefore also with one another. The same result holds for the points of $\overline{\text{Cf.}\,(m+n)_3}$ denoted by primed numerals, if one takes into account the assumption that Cf. m_3 is connected. Finally, since the segments $21'$, $31'$, $2'1$, $3'1$ mediate a transition between the points of the first and second categories, it is clear that all points of $\overline{\text{Cf.}\,(m+n)_3}$ are connected. Q.E.D.

Now consider a connected $(m+n)$-gon, if one exists, whose sides and vertices are the lines and points of $\overline{\text{Cf.}\,(m+n)_3}$. It is clear that in the scheme of this $(m+n)$-gon there are transitions from primed to unprimed numerals and conversely. But altogether there exist only the four transition segments $21'$, $31'$, $2'1$, $3'1$, and since $21'$ and $31'$ belong to the same configuration-line, only one of these two segments can occur as a side of the $(m+n)$-gon; for the same reason, only one of the segments $2'1$, $3'1$ can occur as a side. It follows from all this that exactly one of the segments $21'$ and $31'$, and likewise exactly one of the segments $2'1$ and $3'1$, forms a side of the $(m+n)$-gon; hence the segments 23 and $2'3'$ (since they also belong to the lines $1', 2, 3$ and $1, 2', 3'$) are certainly not sides of the $(m+n)$-gon. Thus we have proved:

(1) There exist connected configurations which contain segments that cannot occur as a segment in any simple closed polygonal path containing all lines and points of the configuration.

Suppose that $\overline{\text{Cf.}\,(m+n)_3}$ can be traversed in one path, and that the segment $21'$ occurs as a polygon side. Then the polygon plainly has the following scheme:

$$\begin{array}{ccc} \text{primed numerals} & \text{unprimed numerals} & \text{primed numerals} \\ \cdots\ 1' & 2\ \cdots\ 1 & \cdots \end{array}$$

and it is then easy to see that the scheme

$$\begin{array}{c} \text{unprimed numerals} \\ 2\ \cdots\ 1 \end{array}$$

represents an n-gon containing all lines and points of Cf. n_3, because in Cf. n_3 the points 2 and 1 are joined. Hence:

(2) If $\overline{\text{Cf.}\,(m+n)_3}$ *can be traversed in a simple closed polygonal path which contains the segment* $21'$*, then* Cf. n_3 *can be traversed in a simple path which contains the segment 21.*

We now suppose that our configuration Cf. n_3 already has the kind described under (1), and specifically that the segment 21 cannot occur in any simple polygonal path completely traversing Cf. n_3. Then it follows from (2) that $\overline{\text{Cf.}\,(m+n)_3}$ also cannot be traversed in a simple polygonal path which contains the segment $21'$. But, as shown above, such a polygonal path cannot contain the segment 23 either. Thus we have proved:

(3) The existence of connected configurations which contain two segments lying on the same line, neither of which is contained in any simple polygonal path traversing the whole configuration.

We may now again suppose that our configuration Cf. n_3 already has the property specified in (3), and that neither of the segments 21 and 31 can be contained in a simple polygonal path encompassing the whole configuration. A simple polygonal path which runs through the whole configuration Cf. $(m+n)_3$ could then contain neither the segment 23 nor, by (2), either of the segments $21'$ and $31'$. Such a polygonal path would therefore contain no segment of the line $1', 2, 3$, which is a contradiction.

Thus the configuration Cf. $(m+n)_3$ can now no longer be traversed in one simple path, and consequently the existence of connected configurations Cf. n_3 for which *Kantor*'s theorem is not valid has been proved.

At the same time the proof contains a procedure for constructing configurations of the described kind. It is easy to see that by repeated application of the procedure one can also produce configurations which are connected and yet cannot be traversed in fewer than k paths, however large the natural number k may be. A simple check of the method, together with the consideration that configurations exist for every $n \geq 7$, shows that for every $n \geq (3k-2) \cdot 7$ configurations of the stated kind must exist.